FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XI

JEE (Advanced)-2021

PAPER –1

TEST DATE: 26-09-2021

Time Allotted: 3 Hours **Maximum Marks: 198**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

  Section-A (01 – 06, 19 – 24, 37– 42) contains **18 multiple** choice questions which have **ONLY ONE CORRECT ANSWER**. Each question carries **+3 marks** for correct answer and **–1 mark** for wrong answer.

  Section-A (07 – 12, 25 – 30, 43 – 48) *this section contains* ***18 multiple*** *choice questions.*
  Each question has FOUR options. ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct answer(s).*
  For each question, choose the option(s) corresponding to (all) the correct answer(s)
  Answer to each question will be evaluated according to the following marking scheme:

  Full Marks **: +4** If only (all) the correct option(s) is (are) chosen:
  Partial Marks **: +3** If all the four options are correct but ONLY three options are chosen;
  Partial Marks **: +2** If three or more options are correct but ONLY two options are chosen and both of which are correct;
  Partial Marks **: +1** If two or more options are correct but ONLY one option is chosen and it is a correct option;
  Zero Marks **: 0** If none of the options is chosen (i. e. the question is unanswered);
  Negative Marks **: –2** In all other cases.

  Section-C (13 – 18, 31– 36, 49 – 54) *contains* ***18 Numerical*** *answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***0 marks*** *for wrong answer.*

Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A ball of radius R is uniformly charged with the volume charge density ρ. The electric flux across the ball's section formed by the plane located at a distance a(<R) from the centre of the ball will be

(A) $\dfrac{\pi\rho a(R^2 - a^2)}{\varepsilon_0}$

(B) $\dfrac{\pi\rho a(R^2 - a^2)}{3\varepsilon_0}$

(C) $\dfrac{\pi\rho a(R^2 - a^2)}{2\varepsilon_0}$

(D) zero

2. An object is projected vertically upward in air from ground. It explodes at the top most point of its trajectory into three identical fragments. One of the fragments comes straight down to ground in time t_1 while the other two come to ground at a time t_2 after explosion. Height at which the explosion occurred is (Take acceleration due to gravity = g and neglect air resistance)

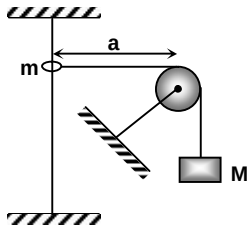
(A) $\dfrac{gt_1t_2(t_2 + 2t_1)}{(t_1 + 2t_2)}$

(B) $\dfrac{gt_1t_2}{2}$

(C) $\dfrac{gt_1t_2(2t_2 + t_1)}{2(t_2 + 2t_1)}$

(D) $\dfrac{gt_1t_2(2t_2 + t_1)}{4(t_2 + 2t_1)}$

3. A ring of mass m slides on a smooth vertical rod. A light string is attached to the ring and passes over a smooth fixed pulley distant 'a' from the rod. At the other end of the string a block of mass M(>m) is attached. The ring is held on a level with the pulley and released. It first comes to rest after falling a distance equal to



(A) $\dfrac{2mMa}{M^2 - m^2}$

(B) $\dfrac{mMa}{M^2 - m^2}$

(C) $\dfrac{mMa}{2(M^2 - m^2)}$

(D) $\dfrac{3mMa}{M^2 - m^2}$

4. A particle starts from rest and moves in a straight line with initial acceleration a_0. The acceleration reduces continuously to half of its value in every t_0 seconds. The terminal velocity of the particle is

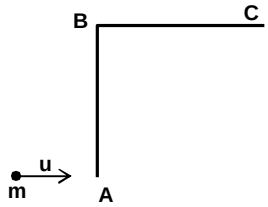
(A) $\frac{a_0 t_0}{2\ell n2}$

(B) $\frac{2a_0 t_0}{\ell n2}$

(C) $\frac{a_0 t_0}{\ell n2}$

(D) $\frac{a_0 t_0}{3\ell n2}$

5. Two rods AB and BC each of mass 'm' and length 'ℓ' are connected rigidly at B and are perpendicular to each other. They lie on a frictionless horizontal surface. A particle of mass 'm' moving on the surface perpendicular to rod AB collides at end A with speed 'u'. If the particle stops after collision, find the angular speed of the connected rod after collision.



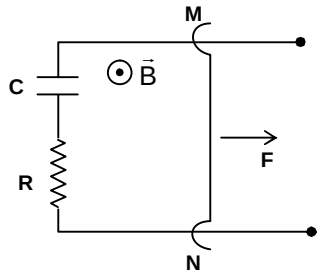
(A) $\frac{9u}{5\ell}$

(B) $\frac{u}{\ell}$

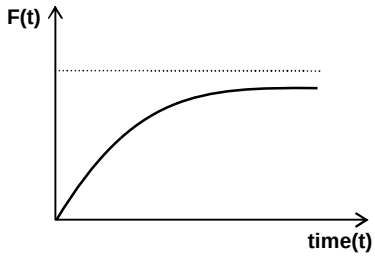
(C) $\frac{5u}{9\ell}$

(D) $\frac{8u}{5\ell}$

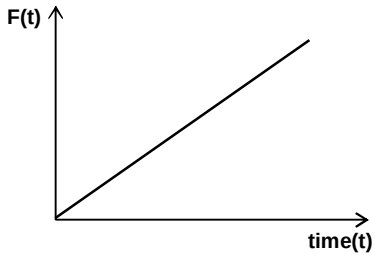
6. A straight perfectly conducting wire MN is connected to a pair of horizontal rails of zero resistance. One side of the rails is connected to a capacitor of capacitance C and a resistor of resistance R. There is a uniform magnetic field B directed normally outwards to the plane of the paper in the entire region. The wire MN is moved with a uniform velocity v by applying a time varying force F. If F(t), I(t) and P(t) denotes the instantaneous value of the applied force, the current in the circuit and the rate of energy stored in the capacitor, then the correct graph will be



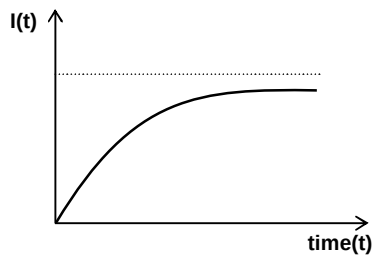
(A)



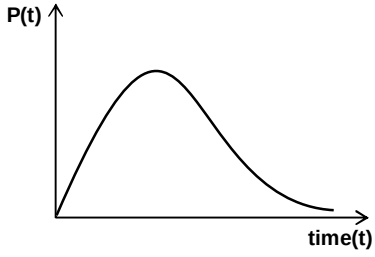
(B)



(C)



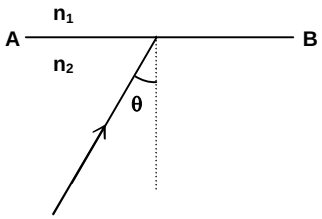
(D)

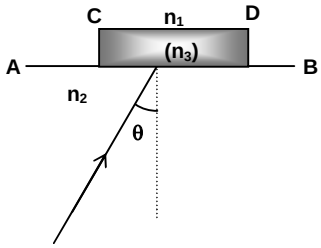


(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

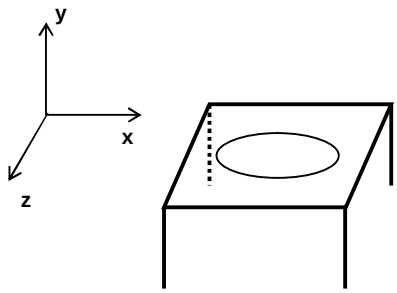
7. An interface AB divides two media of refractive indices n_1 and $n_2(>n_1)$ respectively. A ray of light is incident on this interface from medium of refractive index n_2 at an angle θ which is slightly greater than the critical angle for the pair of two media. Now keeping the direction of incident ray fixed, a parallel long slab of refractive index n_3 is placed on the surface AB as shown, which of the following statement(s) is/are correct.





(A) total internal reflection occurs at AB for $n_3 < n_1$.
(B) total internal reflection occurs at AB for $n_3 > n_1$.
(C) total internal reflection occurs at CD for $n_3 > n_1$.
(D) the ray will return back to medium of refractive index n_2 for all values of n_3.

8. A uniform conducting ring of mass π kg and radius 1 m is kept on a smooth horizontal table. A uniform but time varying magnetic field $\vec{B} = (\hat{i} + t^2\hat{j})$ tesla is present in the region (where t is the time in sec). Resistance of the ring is 2 Ω. Then choose the correct option(s). (Take g = 10 m/s^2), (here y-axis is in the vertical direction)



(A) Time when ring starts toppling is $\frac{10}{\pi}$ sec.

(B) Time when ring starts toppling is $\frac{5}{\pi}$ sec.

(C) Heat generated in the ring till the ring starts toppling is $\frac{2}{3\pi}\times 10^3$ J.

(D) Heat generated in the ring till the ring starts toppling is $\frac{3}{2\pi}\times 10^3$ J.

9. A radio nuclide with decay constant λ is being produced in a reactor at constant rate of p nuclei per sec. At t = 0 the number of radio nuclide was zero. During each decay, E_0 energy is released. If the production of radionuclide is started at t = 0, then

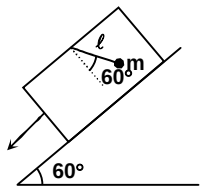
(A) rate of release of energy as a function of time t is $pE_0\left(1-e^{-\lambda t}\right)$

(B) rate of release of energy as a function of time t is $pE_0\left(1+e^{-\lambda t}\right)$

(C) total energy released in time t is $pE_0\left[t-\frac{1}{\lambda}\left(1-e^{-\lambda t}\right)\right]$

(D) total energy released in time t is $pE_0\left[t+\frac{1}{\lambda}\left(1-e^{-\lambda t}\right)\right]$

10. A large heavy box is sliding without friction on an inclined plane of the inclination 60°. A pendulum is suspended from the roof of the box. The mass of the bob is m and length of the string is ℓ. The bob is released from rest when the string is straight and makes angle 60° with the normal to the inclined plane. Then, at the instant when the string is along the normal to the inclined plane.



(A) Speed of the bob relative to the box is equal to $\sqrt{g\ell}$

(B) Speed of the bob relative to the box is equal to $\sqrt{\frac{g\ell}{2}}$

(C) Tension in the string is equal to mg

(D) Tension in the string is equal to $\frac{mg}{2}$

11. The air column in an organ pipe of length L closed at one end and open at the other end, is vibrating in second overtone. Neglect the end correction. Then choose the correct option(s).

(A) Inside the pipe minimum distance from the closed end where pressure remains constant with time is $\frac{L}{5}$.

(B) Wavelength of the wave propagating in the pipe is $\frac{4L}{5}$

(C) The ratio of pressure amplitude at the middle of the pipe to the maximum pressure amplitude in the pipe is $\frac{1}{\sqrt{2}}$

(D) The ratio of displacement amplitude to the maximum displacement amplitude of air particle in the pipe at a distance $\frac{L}{3}$ from the closed end is $\frac{1}{2}$

12. If Planck constant h, Gravitational constant G and viscosity of a fluid η are considered as fundamental quantities, then dimensions of mass M and length L will be

(A) $[M]=\left[h^{\frac{5}{9}}G^{\frac{-1}{3}}\eta^{\frac{1}{9}}\right]$

(B) $[M]=\left[h^{\frac{-1}{3}}G^{\frac{5}{9}}\eta^{\frac{1}{9}}\right]$

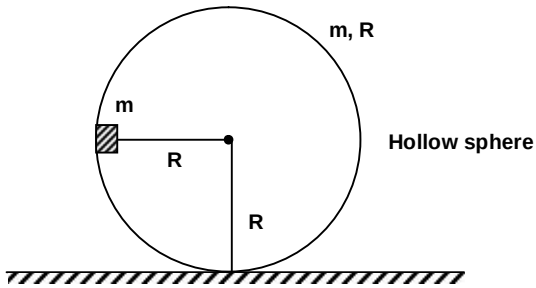
(C) $[L]=\left[h^{\frac{1}{3}}G^{0}\eta^{-\frac{1}{3}}\right]$

(D) $[L]=\left[h^{\frac{5}{9}}G^{0}\eta^{-\frac{1}{3}}\right]$

SECTION – C
(Numerical Answer Type)

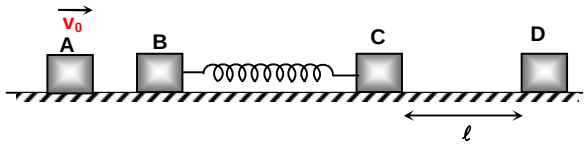
This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. Block of mass 'm' is released from height 'R' inside a smooth hollow sphere of same mass 'm' and radius R. Friction on ground is sufficient to prevent slipping of hollow sphere. The speed of hollow sphere is $\sqrt{\frac{kgR}{10}}$, when block reaches lowest point of sphere. Find the value of k.



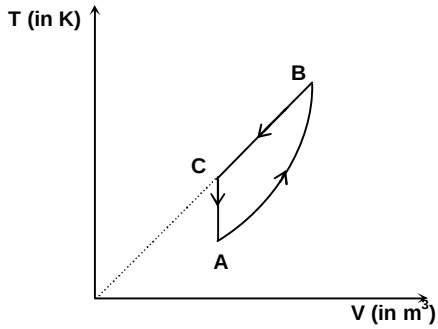
14. The image of a fixed point object inside water $\left(\mu_w = \frac{4}{3}\right)$ is formed in air by a thin convex lens placed just outside the water surface, at a distance of 15 cm from the lens. When this lens is put just inside the water surface, the image of the object is formed in air at a distance of 60 cm from the lens. Find the focal length (in cm) of the lens in air. The refractive index of the material of the lens is 1.5.

15. Two identical blocks B and C are connected by a light spring of force constant k. They are placed on a smooth horizontal surface. A third identical block D is placed on the surface at a distance ℓ to the right of block C as shown.



Now a fourth identical block A moving on the surface with a velocity v_0 makes a head on elastic collision with the block B. All the collisions between the blocks are elastic and head on. If the minimum value of ℓ for which the block D moves with a velocity v_0 subsequently is $n\pi v_0\sqrt{\frac{2m}{k}}$, find the value of n. (Take mass of each block is m)

16. One mole of a diatomic ideal gas $\left(C_V = \frac{5R}{2}\right)$ is made to undergo a quasi-static cyclic thermodynamics process. During process A $\rightarrow$ B, the temperature T of the gas varies with its volume V according to the equation $T = kV^2$, where k is a constant. During the whole cycle, maximum pressure of the gas is two times of the minimum pressure. If the gas absorbs 0.36 kJ of heat during process A $\rightarrow$ B, how much heat (in kJ) does the gas liberate during the process B $\rightarrow$ C.



17. The maximum range of a shell fired from a fixed gun is R = 40 m. Now the gun is mounted on a platform that moves horizontally with a constant velocity 20 m/s. What would be the maximum range (in m) of a shell fired from the gun mounted on the platform? (Given $\sqrt{3} = 1.732$, g = 10 m/s^2)

18. Consider a vernier callipers in which 1 cm on the main scale is divided into 10 equal divisions. The vernier scale of the callipers has 10 equal divisions that correspond to 11 main scale divisions. When nothing is kept between the jaws of the callipers and the jaws touch each other the zero of the vernier scale is slightly left to the zero of the main scale and fourth division of the vernier scale coincides with a main scale division as shown in figure-(i). When thickness of a plate is measured by this vernier callipers the zero of the vernier scale lies between 3.2 cm and 3.3 cm mark of the main scale and third division of the vernier scale coincides with a main scale division as shown in the figure-(ii). The thickness of the plate (in cm) is

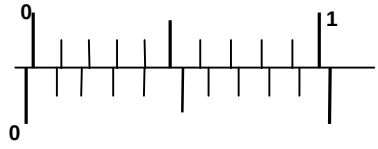


figure-(i)

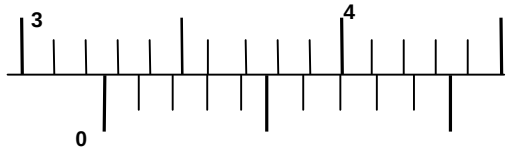


figure-(ii)

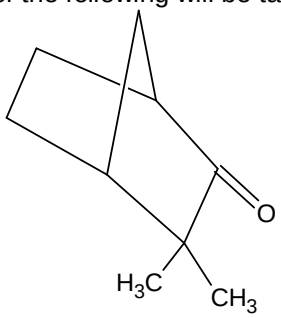
Chemistry

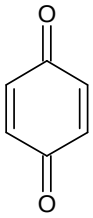
PART – II

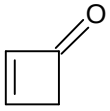
SECTION – A
(One Options Correct Type)

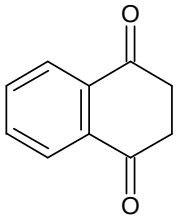
This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

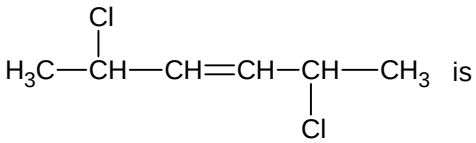
19. Number of geometrical isomers for the molecule
$CH_3 - CH = CH - CH = CH - CH = CH - CH_3$ is
(A) 8
(B) 4
(C) 6
(D) 2

20. Which of the following will be tautomerise?
(A)


(B)


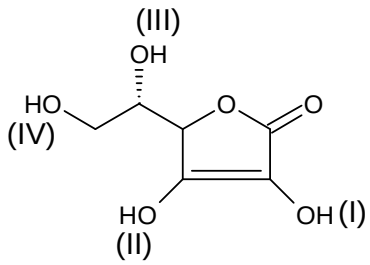
(C)


(D)


21. Total number of stereoisomers for the


is
(A) 6
(B) 8
(C) 4
(D) 16

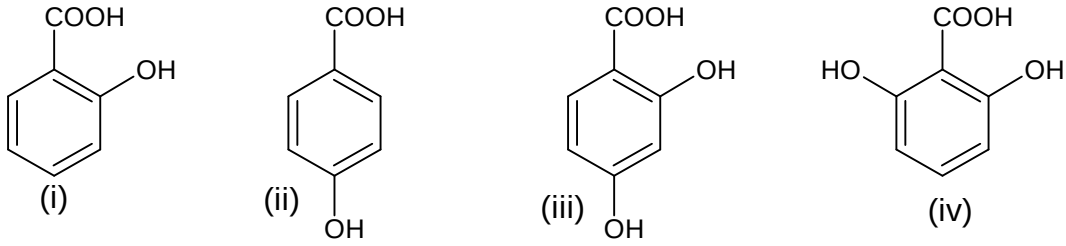
22.



The most acidic hydrogen in the above molecule is

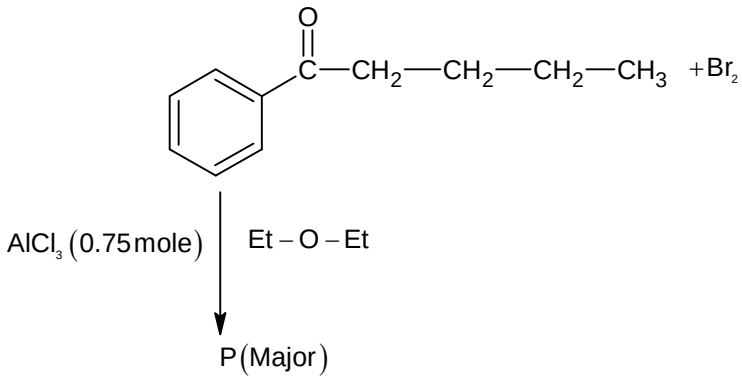
(A) I
(B) II
(C) III
(D) IV

23. The order of pK_a values of the following acid is

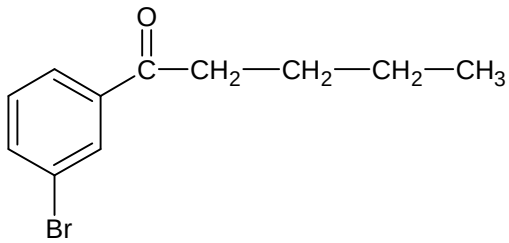


(A) iv > i > iii > ii
(B) iii > iv > i > ii
(C) ii > i > iii > iv
(D) ii > iii > i > iv

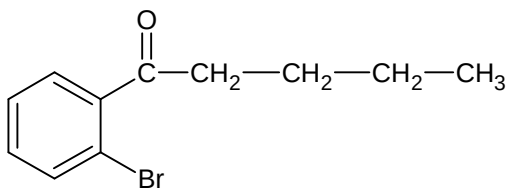
24.

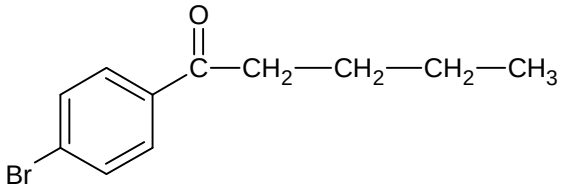


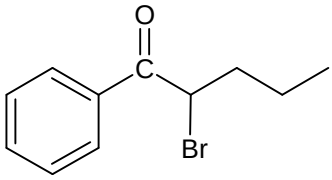
(A)



(B)



(C) 

(D) 

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

25. Which of the following shows co-ordination isomerism?
    (A) $\left[Ag(NH_3)_2\right][AgCl_2]$
    (B) $\left[Pt(NH_3)_4\right][PtCl_4]$
    (C) $\left[Co(en)_3\right]\left[Cr(CN)_6\right]$
    (D) $\left[Pt(NH_3)_4\right][PtCl_6]$

26. Among the following paramagnetic complex is/are
    (A) $\left[Ni(CO)_4\right]$
    (B) $\left[Cu(NH_3)_4\right]^{2+}$
    (C) $\left[Fe(CN)_6\right]^{3-}$
    (D) $\left[Fe(CN)_6\right]^{4-}$

27. Among the following yellow colour sulphide is/are
    (A) CdS
    (B) As_2S_3
    (C) SnS
    (D) PbS

28. Which of the following on heating gives N_2?
    (A) NH_4NO_3
    (B) NH_4NO_2
    (C) $(NH_4)_2Cr_2O_7$
    (D) $(NH_4)_3PO_4$

29. Which of the following reaction is/are correct?
    (A) $CsF + IF_7 \longrightarrow Cs^+\left[IF_8\right]^-$
    (B) $I_2O_5 + H_2O \longrightarrow HIO_4$
    (C) $2XeF_6 + SiO_2 \longrightarrow 2XeOF_4 + SiF_4$
    (D) $6XeF_4 + 12H_2O \longrightarrow 2XeO_3 + 4Xe + 24HF + O_2$

30. Which of the following is/are act as buffer solution?

(A) aqueous solution of Borax

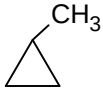
(B) aq. $CH_3-\overset{\overset{O}{\|}}{C}-OH$ solution

(C) $CH_3-\overset{\overset{O}{\|}}{C}-OH$ + NaOH [2 : 1 molar ratio]

(D) $H_3PO_4 + NaOH\ [1 : 2.5 \text{ molar ratio}]$

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

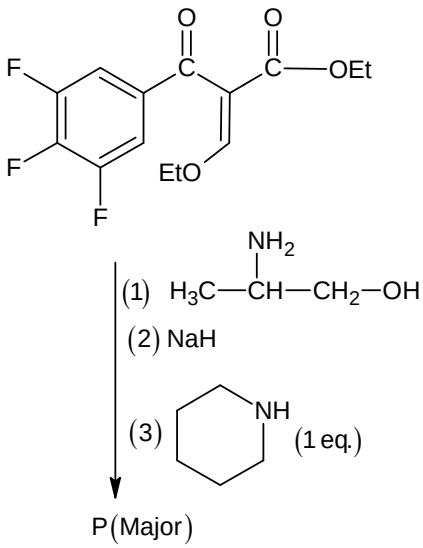
31. Total number of Dihalogen derivative of  is x (including stereoisomers) then value of x/5 is

32. 1.2 g carbon is burnt completely in oxygen (limited supply) to product CO and CO_2 This mixture of gases is treated with solid I_2O_5. The liberated iodine required 120 ml of 0.1 M hypo solution for complete titration. The % of carbon converted into CO is x. Then value of x/4 is

33. Number of B – O – B bond in borax is x then value of x/4 is?

34. For a fixed amount of real gas when a graph of Z vs. P was plotted at very high pressure slope was observed to be $0.01\,atm^{-1}$. At the same temperature (high pressure) if a graph is plotted between PV vs. P then for 2 mole of the gas Y intercept is found to be 40 atm – litre. Excluded volumes in litres for 9 mole of the real gas is x. Then the value of x is.

35. Number of stereo isomers for the following complex $[Ma_2b_2cd]$ is x (where a, b, c, d are monodentate ligand) then value of x/5 is?

36. 

If unsaturation factor in major product P is x then value of x/4 is?

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. Let f(x) be a derivable function at x = 2 and $\lim_{h\to 0}\frac{f(2+h)}{\sin h}=3$, then $\left|\frac{f(2)+f'(2)}{f'(2)-f(2)}\right|$ is

(A) 1
(B) 0
(C) 2
(D) 3

38. If $\left|\log_{\sqrt{2}} 30-\left|\log_2 9+\left|\log_4 9-\left|\log_{1/2} 5\right|\right|\right|\right| = x$, then the smallest integer greater than or equal to x is

(A) 5
(B) 6
(C) 4
(D) 7

39. If k be the minimum value of expression $(\cos^2\alpha-2\cos\alpha)\sec^2\beta+9\operatorname{cosec}^2\beta+5\sec^2\beta$ where $\alpha\in[0,\pi]$ and $\beta\in\left(0,\frac{\pi}{2}\right)$, then k is

(A) 20
(B) 25
(C) 24
(D) 29

40. If $f(x) = x^2 + ax + b$ where a, b $\in$ R and f(f(x)) = 0 has roots 1 and 2. If k be the value of f(x) at x = 0, then k is

(A) $\frac{3}{2}$
(B) 2
(C) $-\frac{3}{2}$
(D) –2

41. If $y=\tan^{-1}\left(\frac{4x}{1+5x^2}\right)+\tan^{-1}\left(\frac{2+3x}{3-2x}\right)$ where $x\in\left(0,\frac{2}{3}\right)$. If $\frac{dy}{dx}=\frac{k}{1+\lambda x^2}$ where k and $\lambda\in$ N. Then $\frac{\lambda}{k}$ is

(A) 5
(B) 7
(C) 9
(D) 6

42. If α and β be the largest and smallest value of $\frac{y}{x}$, where (x, y) is real number satisfying $(x-3)^2 + (y-3)^2 = 6$, then $|\alpha + \beta|$ is
(A) 2
(B) 6
(C) 9
(D) 7

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

43. Let a, b, c denotes side lengths of ΔABC. If a, b, c are the roots of equation $8x^2 + (\lambda + 2)x^2 - (2k + \lambda)x - 27 = 0$, such that $\lambda^2 + 2\lambda(k + 1) + 4k = 2^3 \cdot 3^5$, then which of the following is/are correct

(A) circumradius of ΔABC is $\frac{\sqrt{3}}{2}$

(B) distance between orthocentre and side AB is $\frac{\sqrt{3}}{4}$

(C) distance between orthocentre and circumcentre of ΔABC is $\frac{\sqrt{3}}{4}$

(D) distance between orthocentre and side BC is $\frac{\sqrt{3}}{2}$

44. The value of $\sum_{k=1}^{\infty} \frac{6^k}{(3^k - 2^k)(3^{k+1} - 2^{k+1})}$ can be equal to

(A) $\lim_{x \to 0} \frac{1 - \cos(2x)}{x^2}$

(B) $\lim_{n \to \infty} \left(\sin\left(\frac{\pi}{2n}\right) \times \sin\left(\frac{2\pi}{2n}\right) \right) \times ..... \times \sin\left(\frac{(n-1)\pi}{n}\right)^{1/n}$

(C) $\frac{\pi \ln 2}{\int_{\pi/2}^{0} \ln|\cos(2x)|\,dx}$

(D) $\frac{2}{\pi \ln 2} \int_{0}^{\pi/4} \ln(1 + \tan x)\,dx$

45. Which of the following is/are correct?
(A) If A and B are two square matrices of order 3 and A is a non-singular matrix such that AB = O, then B must be a null matrix
(B) It A, B, C are three square matrices of order 2 and det(A) = 2, det(B) = 3, det(C) = 4, then the value of det(3 ABC) is 216
(C) It A is a square matrix of order 3 and det(A) = $\frac{1}{2}$, then det(adj A^{-1}) is 8
(D) Every skew symmetric matrix is singular

46. If $^{100}C_{50}$ can be prime factorised as $2^x \cdot 3^y \cdot 5^z \cdot 7^k$ ....., where x, y, z, k, ... are non-negative integers, then correct statement is/are
(A) $x < y$
(B) $z < k$
(C) $x + k = y + z - 1$
(D) $z + k = 0$

47. If k be the remainder when $\left(\sum_{r=1}^{5} {}^{20}C_{(2r-1)}\right)^5$ is divided by 11, then k is smaller than or equal to is/are
(A) 4
(B) 3
(C) 2
(D) 1

48. If $z = \cos(0.001) + i\sin(0.001)$ be a complex number. Then the possible positive integer value of k such that z^k has an imaginary part which exceeds 0.5 is/are
(A) 522
(B) 524
(C) 526
(D) 529

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. If k be the set of all continuous real valued functions which are solutions to $(f(x))^2 = 100 + \int_0^x (f(t)f'(t) - f(t) - f'(t) - 1)\,dt$, then the value of $\frac{1}{|k|}\sum_{f(x)\in k} |f(100)|$ is

50. $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar unit vectors inclined at an angle $\theta \left(\leq \frac{\pi}{2}\right)$ to each other. If the volume of tetrahedron formed by these vectors is $\frac{1}{\sqrt{360}}$, then the value of $|3\cos^2\theta - 2\cos^3\theta|$ is

51. If f and g are differentiable functions such that
$xg(f(x))f'(g(x))g'(x) = f(g(x))g'(f(x))f'(x)$ for all real x. Also f is non-negative and g is positive. If $\int_0^p f(g(x))\,dx = \frac{1}{2} - \frac{e^{-2p}}{2}$ for all real p and g(f(0)) = 1, If the value of g(f(4)) is e^{-k}, where $k \in N$. Then k is

52. Let the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, contains the circle $(x - 1)^2 + y^2 = 1$ and has least area. If $a^2 + b^2 = 12k$, where $k \in N$, then k is

53. A straight line L with the slope $\frac{3}{4}$ touches a circle whose radius is 5 units and centre lies on the x-axis. If the length of this tangent from a point on the x-axis is $\frac{p}{q}$ where p, q $\in$ N, then the least value of p + q is

54. The point of intersection of the plane $\vec{r} \cdot \left(3\hat{i} - 5\hat{j} + 2\hat{k}\right) = 6$ with the straight line passing through the origin and perpendicular to the plane 2x – y – z = 4, is (α, β, γ), then the value of $\left|\frac{2\alpha - 3\beta + \gamma}{5}\right|$ is